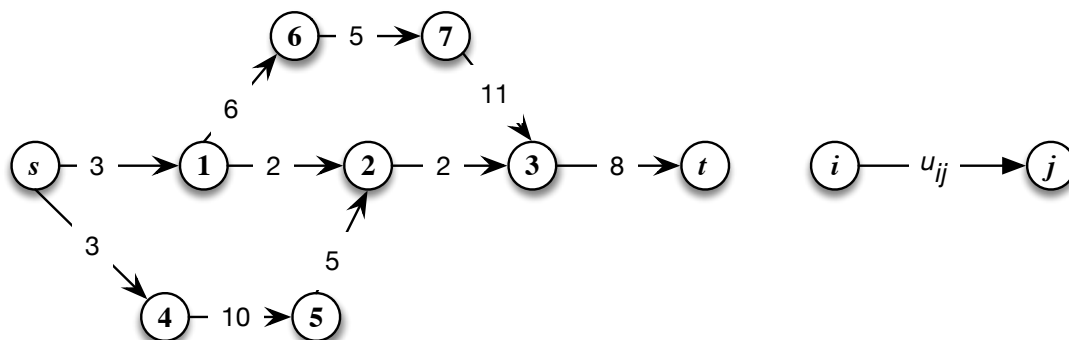


First part : Closed book

Total score: 21 pts. Minimum score to pass: 11 pts. Tentative scores: 1) 4 pts, 2) 5 pts, 3) 5 pts, 4) 3 pts.

1. Determine the maximum flow in the following graph, starting from the null flow ($x_{ij} = 0, \forall (i, j) \in A$). Apply the Edmonds and Karp algorithm. Report the intermediate steps and the final minimum capacity cut.

Does the maximum flow changes if the capacity of arc (1,2) increases? Why? What about arc (2,3)?



2. Consider the following linear programming problem:

$$\begin{aligned}
 \max \quad & x_1 + x_2 \\
 & -4x_1 + 8x_2 \leq 14 \\
 & 4x_1 - x_2 \leq 7 \\
 & 2x_2 \leq 5 \\
 & x_1 - x_2 \leq 1
 \end{aligned}$$

Consider the point $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Determine the active constraints in x . Write the condition of existence of a growing feasible direction (RP) and the optimality condition (RD). Give the two geometric representations of RP and RD. Check whether the direction $\bar{\xi} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ is actually a growing feasible direction. In the positive case, determine the moving step along $\bar{\xi}$ and the new feasible point.

3. Consider the following Integer Linear Programming problem:

$$\begin{aligned}
 \max \quad & -2x_1 + x_2 \\
 & -2x_1 + 4x_2 \leq 7 \\
 & 6x_1 + 8x_2 \leq 27 \\
 & x_1, x_2 \geq 0, \text{ integer}
 \end{aligned}$$

Add the slack variables (x_3 and x_4) to the first and the second constraints, and consider the optimal base $B = \{2, 4\}$ of the LP relaxation. Generate the Gomory cuts associated with the fractional components of the solution. Justify your answer and give the geometric representation of the feasible region and of the cut(s).

4. Consider the feasible region $F \subset \mathbb{Z}^n$. Give the definition of convex hull of F ($\text{conv}(F)$). What is the most important property of $\text{conv}(F)$ when F contains only integer coordinate points? For $F = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}$ find the inequalities defining its convex hull.

AMPL questions: 4 points.

in the rear of this paper

Write the definition in the ampl modelling language of the following:

- two parameters a and b , suppose that $a < b$;
- a set I of all the integers between a and b ;
- a binary variable x_i with $i \in I$;
- a parameter c_i with $i \in I$;

Then using the defined variables and parameters write in the AMPL modeling language the following constraint:

$$\sum_{i \in I, i \leq c_j} x_i \leq 5 \qquad \forall j \in I$$

ANSWER IN THE SPACE BELOW